

Администрирование локальных сетей

Настройка VPN

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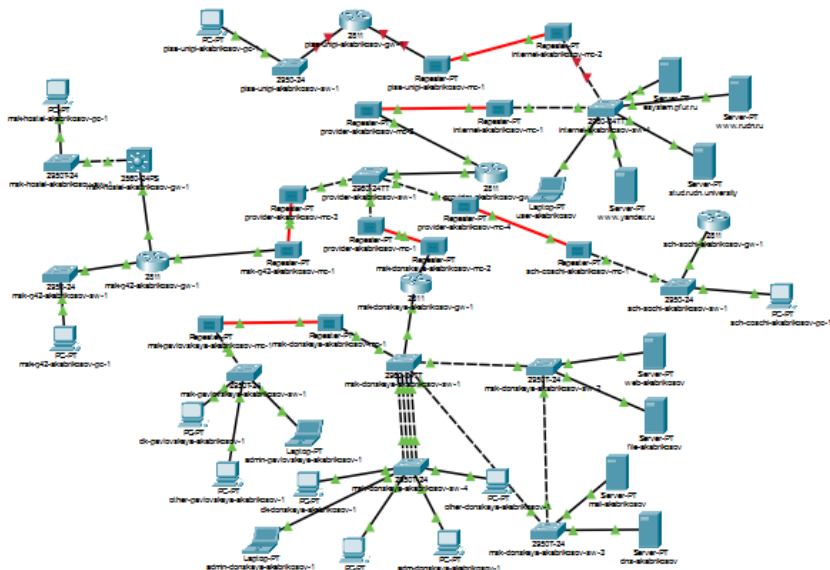
24 мая 2026

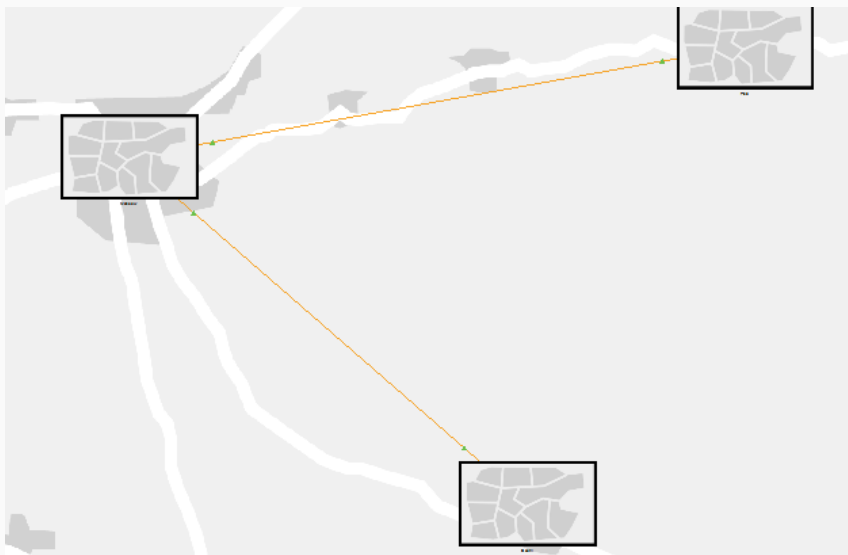
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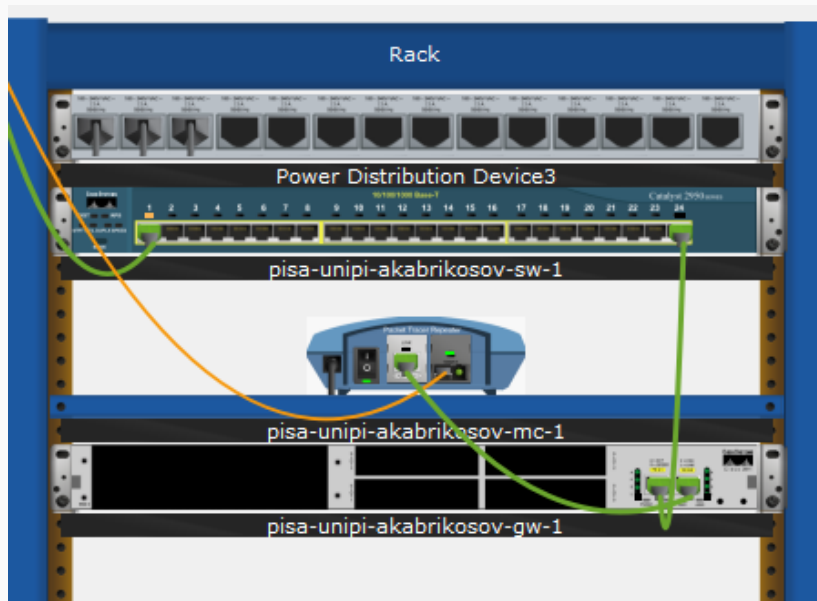
Цели и задачи работы

Получить навыки настройки VPN-туннеля через незащищённое Интернет-соединение.

Размещение оборудования







Первоначальная настройка устройств

Настройка маршрутизатора

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#host pisa-unipi-akabrikosov-gw-1
pisa-unipi-akabrikosov-gw-1(config)#line vty 0 4
pisa-unipi-akabrikosov-gw-1(config-line)#pass cisco
pisa-unipi-akabrikosov-gw-1(config-line)#login
pisa-unipi-akabrikosov-gw-1(config-line)#exit
pisa-unipi-akabrikosov-gw-1(config)#
pisa-unipi-akabrikosov-gw-1(config)#line console 0
pisa-unipi-akabrikosov-gw-1(config-line)#pass cisco
pisa-unipi-akabrikosov-gw-1(config-line)#login
pisa-unipi-akabrikosov-gw-1(config-line)#exit
pisa-unipi-akabrikosov-gw-1(config)#
pisa-unipi-akabrikosov-gw-1(config)#enab sec cisco
pisa-unipi-akabrikosov-gw-1(config)#serv pass
pisa-unipi-akabrikosov-gw-1(config)#user admin priv 1 sec cisco
pisa-unipi-akabrikosov-gw-1(config)#ip domain-name unipi.edu
pisa-unipi-akabrikosov-gw-1(config)#crypto key gen rsa
The name for the keys will be: pisa-unipi-akabrikosov-gw-1.unipi.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

pisa-unipi-akabrikosov-gw-1(config)#line vty 0 4
*Mar 1 0:6:2.748: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:6:2.748: %SSH-5-ENABLED: SSH 1.5 has been enabled
pisa-unipi-akabrikosov-gw-1(config-line)#trans in ssh
pisa-unipi-akabrikosov-gw-1(config-line)#exit
pisa-unipi-akabrikosov-gw-1(config)#
```

Настройка интерфейсов маршрутизатора

```
-----
pisa-unipi-akabrikosov-gw-1(config)#int f0/0
pisa-unipi-akabrikosov-gw-1(config-if)#no sh

pisa-unipi-akabrikosov-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

pisa-unipi-akabrikosov-gw-1(config-if)#exit
pisa-unipi-akabrikosov-gw-1(config)#int f0/0.401
pisa-unipi-akabrikosov-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.401, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.401, changed state to up

pisa-unipi-akabrikosov-gw-1(config-subif)#enc dot1q 401
pisa-unipi-akabrikosov-gw-1(config-subif)#ip addr 10.131.0.1 255.255.255.0
pisa-unipi-akabrikosov-gw-1(config-subif)#descr unipi-main
pisa-unipi-akabrikosov-gw-1(config-subif)#exit
pisa-unipi-akabrikosov-gw-1(config)#int f0/1
pisa-unipi-akabrikosov-gw-1(config-if)#no sh

pisa-unipi-akabrikosov-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

pisa-unipi-akabrikosov-gw-1(config-if)#ip addr 192.0.2.20 255.255.255.0
pisa-unipi-akabrikosov-gw-1(config-if)#descr internet
pisa-unipi-akabrikosov-gw-1(config-if)#exit
pisa-unipi-akabrikosov-gw-1(config)#ip route 0.0.0.0 0.0.0.0 192.0.2.1
pisa-unipi-akabrikosov-gw-1(config)#
```

Настройка коммутатора

```
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#line vty 0 4
Switch(config-line)#pass cisco
Switch(config-line)#login
Switch(config-line)#exit
Switch(config)#line cons 0
Switch(config-line)#pass cisco
Switch(config-line)#login
Switch(config-line)#exit
Switch(config)#enab sec cisco
Switch(config)#serv pass
Switch(config)#user admin priv 1 sec cisco
Switch(config)#ip domain-name unipi.edu
Switch(config)#crypto key gen rsa
% Please define a hostname other than Switch.
Switch(config)#host pisa-unipi-akabrikosov-sw-1
pisa-unipi-akabrikosov-sw-1(config)#crypto key gen rsa
The name for the keys will be: pisa-unipi-akabrikosov-sw-1.unipi.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

pisa-unipi-akabrikosov-sw-1(config)#line vty 0 4
*Mar 1 0:12:3.458: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:12:3.458: %SSH-5-ENABLED: SSH 1.5 has been enabled
pisa-unipi-akabrikosov-sw-1(config-line)#trans in ssh
pisa-unipi-akabrikosov-sw-1(config-line)#exit
pisa-unipi-akabrikosov-sw-1(config)#
```

Настройка VLAN и портов

```
pisa-unipi-akabrikosov-sw-1(config)#
pisa-unipi-akabrikosov-sw-1(config)#int f0/24
pisa-unipi-akabrikosov-sw-1(config-if)#sw mode trunk

pisa-unipi-akabrikosov-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

pisa-unipi-akabrikosov-sw-1(config-if)#exit
pisa-unipi-akabrikosov-sw-1(config)#int f0/1
pisa-unipi-akabrikosov-sw-1(config-if)#sw mode acc
pisa-unipi-akabrikosov-sw-1(config-if)#sw acc vlan 401
% Access VLAN does not exist. Creating vlan 401
pisa-unipi-akabrikosov-sw-1(config-if)#exit
pisa-unipi-akabrikosov-sw-1(config)#vlan 401
pisa-unipi-akabrikosov-sw-1(config-vlan)#name unipi-main
pisa-unipi-akabrikosov-sw-1(config-vlan)#exit
pisa-unipi-akabrikosov-sw-1(config)#int vlan 401
pisa-unipi-akabrikosov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan401, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan401, changed state to up

pisa-unipi-akabrikosov-sw-1(config-if)#no sh
pisa-unipi-akabrikosov-sw-1(config-if)#exit
pisa-unipi-akabrikosov-sw-1(config)#exit
pisa-unipi-akabrikosov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

pisa-unipi-akabrikosov-sw-1#wr mem
Building configuration...
[OK]
pisa-unipi-akabrikosov-sw-1#
```

Настройка оконечного устройства

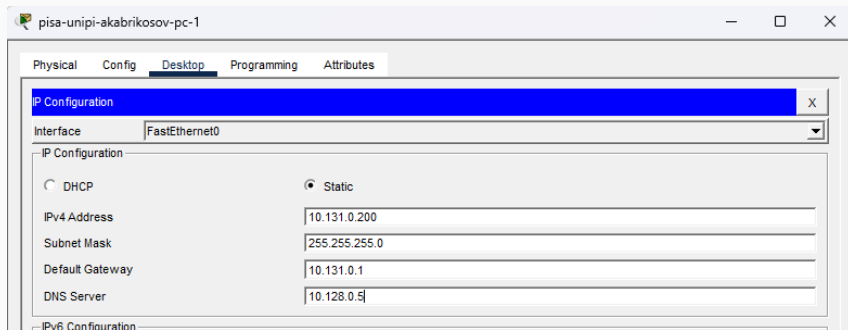


Рис. 8: Настройка статического IP-адреса на ПК сети Университета г. Пиза

Настройка GRE VPN

GRE-туннель на маршрутизаторе Университета г. Пиза

```
pisa-unipi-akabrikosov-gw-1(config)#
pisa-unipi-akabrikosov-gw-1(config)#int tun 0

pisa-unipi-akabrikosov-gw-1(config-if)#
%LINK-5-CHANGED: Interface Tunnel0, changed state to up

pisa-unipi-akabrikosov-gw-1(config-if)#ip addr 10.128.255.254 255.255.255.252
pisa-unipi-akabrikosov-gw-1(config-if)#tun sou f0/1
pisa-unipi-akabrikosov-gw-1(config-if)#tun dest 198.51.100.2
pisa-unipi-akabrikosov-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up

pisa-unipi-akabrikosov-gw-1(config-if)#exit
pisa-unipi-akabrikosov-gw-1(config)#int loop 0

pisa-unipi-akabrikosov-gw-1(config-if)#
%LINK-3-UPDOWN: Interface Loopback0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

pisa-unipi-akabrikosov-gw-1(config-if)#ip addr 10.128.254.5 255.255.255.255
pisa-unipi-akabrikosov-gw-1(config-if)#exit
pisa-unipi-akabrikosov-gw-1(config)#ip route 10.128.254.1 255.255.255.255 10.128.255.253
pisa-unipi-akabrikosov-gw-1(config)#router ospf 1
pisa-unipi-akabrikosov-gw-1(config-router)#router-id 10.128.254.5
pisa-unipi-akabrikosov-gw-1(config-router)#net 10.0.0.0 0.255.255.255 area 0
pisa-unipi-akabrikosov-gw-1(config-router)#exit
```

Рис. 9: Настройка GRE-туннеля и OSPF на маршрутизаторе Университета г. Пиза

GRE-туннель на удалённом маршрутизаторе

```
msk-donskaya-akabrikov-gw-1(config)#int tun 0

msk-donskaya-akabrikov-gw-1(config-if)#
%LINK-5-CHANGED: Interface Tunnel0, changed state to up

msk-donskaya-akabrikov-gw-1(config-if)#ip addr 10.128.255.253 255.255.255.252
msk-donskaya-akabrikov-gw-1(config-if)#tun sou f0/1.4
msk-donskaya-akabrikov-gw-1(config-if)#tun dest 192.0.2.20
msk-donskaya-akabrikov-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up

msk-donskaya-akabrikov-gw-1(config-if)#exit
msk-donskaya-akabrikov-gw-1(config)#
05:02:06: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.5 on Tunnel0 from LOADING to FULL, Loading Done

msk-donskaya-akabrikov-gw-1(config)#int loop 0

msk-donskaya-akabrikov-gw-1(config-if)#
%LINK-3-UPDOWN: Interface Loopback0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

msk-donskaya-akabrikov-gw-1(config-if)#ip addr 10.128.254.1 255.255.255.255
msk-donskaya-akabrikov-gw-1(config-if)#exit
msk-donskaya-akabrikov-gw-1(config)#ip rou 10.128.254.5 255.255.255.255 10.128.255.254
% Ambiguous command: "ip rou 10.128.254.5 255.255.255.255 10.128.255.254"
msk-donskaya-akabrikov-gw-1(config)#ip route 10.128.254.5 255.255.255.255 10.128.255.254
msk-donskaya-akabrikov-gw-1(config)#exit
msk-donskaya-akabrikov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-akabrikov-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-akabrikov-gw-1#
```


Проверка VPN-соединения

Состояние туннеля на маршрутизаторе сети «Донская»

```
msk-donskaya-akabrikosov-gw-1#sh int tun 0
Tunnel0 is up, line protocol is up (connected)
  Hardware is Tunnel
  Internet address is 10.128.255.253/30
  MTU 17916 bytes, BW 100 Kbit/sec, DLY 50000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel source 198.51.100.2 (FastEthernet0/1.4), destination 192.0.2.20
  Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255
  Fast tunneling enabled
  Tunnel transport MTU 1476 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 1
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 26 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    17 packets input, 1372 bytes, 0 no buffer
    Received 0 broadcasts, 0 runs, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 input packets with dribble condition detected
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 0 interface resets
    0 unknown protocol drops
    0 output buffer failures, 0 output buffers swapped out
```

```
msk-donskaya-akabrikosov-gw-1#
```

Таблица маршрутизации сети «Донская»

msk-donskaya-akabrikov-gw-1

Physical Config CLI Attributes

IOS Command Line Interface

```

P - periodic downloaded static route

Gateway of last resort is 198.51.100.1 to network 0.0.0.0

10.0.0.0/8 is variably subnetted, 29 subnets, 4 masks
C    10.128.0.0/24 is directly connected, FastEthernet0/0.3
L    10.128.0.1/32 is directly connected, FastEthernet0/0.3
C    10.128.1.0/24 is directly connected, FastEthernet0/0.2
L    10.128.1.1/32 is directly connected, FastEthernet0/0.2
C    10.128.3.0/24 is directly connected, FastEthernet0/0.101
L    10.128.3.1/32 is directly connected, FastEthernet0/0.101
C    10.128.4.0/24 is directly connected, FastEthernet0/0.102
L    10.128.4.1/32 is directly connected, FastEthernet0/0.102
C    10.128.5.0/24 is directly connected, FastEthernet0/0.103
L    10.128.5.1/32 is directly connected, FastEthernet0/0.103
C    10.128.6.0/24 is directly connected, FastEthernet0/0.104
L    10.128.6.1/32 is directly connected, FastEthernet0/0.104
C    10.128.254.1/32 is directly connected, Loopback0
S    10.128.254.5/32 [1/0] via 10.128.255.254
C    10.128.255.0/30 is directly connected, FastEthernet0/1.5
L    10.128.255.1/32 is directly connected, FastEthernet0/1.5
C    10.128.255.4/30 is directly connected, FastEthernet0/1.6
L    10.128.255.5/32 is directly connected, FastEthernet0/1.6
O    10.128.255.8/30 [110/2] via 10.128.255.2, 00:20:39, FastEthernet0/1.5
    [110/2] via 10.128.255.6, 00:20:39, FastEthernet0/1.6
C    10.128.255.252/30 is directly connected, Tunnel0
L    10.128.255.253/32 is directly connected, Tunnel0
S    10.129.0.0/16 [1/0] via 10.128.255.2
O    10.129.0.0/24 [110/2] via 10.128.255.2, 00:20:49, FastEthernet0/1.5
O    10.129.1.0/24 [110/2] via 10.128.255.2, 00:20:39, FastEthernet0/1.5
O    10.129.128.0/24 [110/3] via 10.128.255.2, 00:20:39, FastEthernet0/1.5
S    10.130.0.0/16 [1/0] via 10.128.255.6
O    10.130.0.0/24 [110/2] via 10.128.255.6, 00:20:49, FastEthernet0/1.6
O    10.130.1.0/24 [110/2] via 10.128.255.6, 00:20:49, FastEthernet0/1.6
O    10.131.0.0/24 [110/1001] via 10.128.255.254, 00:01:43, Tunnel0
198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C    198.51.100.0/28 is directly connected, FastEthernet0/1.4
L    198.51.100.2/32 is directly connected, FastEthernet0/1.4
S*   0.0.0.0/0 [1/0] via 198.51.100.1
```

Состояние туннеля на маршрутизаторе Университета г. Пиза

```
pisa-unipi-akabrikosov-gw-1#sh int tun 0
Tunnel0 is up, line protocol is up (connected)
  Hardware is Tunnel
  Internet address is 10.128.255.254/30
  MTU 17916 bytes, BW 100 Kbit/sec, DLY 50000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel source 192.0.2.20 (FastEthernet0/1), destination 198.51.100.2
  Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255
  Fast tunneling enabled
  Tunnel transport MTU 1476 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 1
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 40 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    23 packets input, 2336 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 input packets with dribble condition detected
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 0 interface resets
    0 unknown protocol drops
    0 output buffer failures, 0 output buffers swapped out

pisa-unipi-akabrikosov-gw-1#
```

Таблица маршрутизации Университета г. Пиза

```
pisa-unipi-akabrikosov-gw-1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 192.0.2.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 20 subnets, 3 masks
O       10.128.0.0/24 [110/1001] via 10.128.255.253, 00:02:25, Tunnel0
O       10.128.1.0/24 [110/1001] via 10.128.255.253, 00:02:25, Tunnel0
O       10.128.3.0/24 [110/1001] via 10.128.255.253, 00:02:25, Tunnel0
O       10.128.4.0/24 [110/1001] via 10.128.255.253, 00:02:25, Tunnel0
O       10.128.5.0/24 [110/1001] via 10.128.255.253, 00:02:25, Tunnel0
O       10.128.6.0/24 [110/1001] via 10.128.255.253, 00:02:25, Tunnel0
S       10.128.254.1/32 [1/0] via 10.128.255.253
C       10.128.254.5/32 is directly connected, Loopback0
O       10.128.255.0/30 [110/1001] via 10.128.255.253, 00:02:25, Tunnel0
O       10.128.255.4/30 [110/1001] via 10.128.255.253, 00:02:25, Tunnel0
O       10.128.255.8/30 [110/1002] via 10.128.255.253, 00:02:25, Tunnel0
C       10.128.255.252/30 is directly connected, Tunnel0
L       10.128.255.254/32 is directly connected, Tunnel0
O       10.129.0.0/24 [110/1002] via 10.128.255.253, 00:02:25, Tunnel0
O       10.129.1.0/24 [110/1002] via 10.128.255.253, 00:02:25, Tunnel0
O       10.129.128.0/24 [110/1003] via 10.128.255.253, 00:02:25, Tunnel0
O       10.130.0.0/24 [110/1002] via 10.128.255.253, 00:02:25, Tunnel0
O       10.130.1.0/24 [110/1002] via 10.128.255.253, 00:02:25, Tunnel0
C       10.131.0.0/24 is directly connected, FastEthernet0/0.401
L       10.131.0.1/32 is directly connected, FastEthernet0/0.401
    192.0.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.0.2.0/24 is directly connected, FastEthernet0/1
L       192.0.2.20/32 is directly connected, FastEthernet0/1
S*    0.0.0.0/0 [1/0] via 192.0.2.1
```

Проверка доступности узла

admin-donskaya-akabrikosov-1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.131.0.200

Pinging 10.131.0.200 with 32 bytes of data:

Request timed out.
Reply from 10.131.0.200: bytes=32 time=1ms TTL=126
Reply from 10.131.0.200: bytes=32 time=2ms TTL=126
Reply from 10.131.0.200: bytes=32 time<1ms TTL=126

Ping statistics for 10.131.0.200:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 1ms

C:\>tracert 10.131.0.200

Tracing route to 10.131.0.200 over a maximum of 30 hops:

  0  4 ms    0 ms    0 ms    10.128.6.1
  1  0 ms    0 ms    0 ms    10.128.255.254
  2  0 ms    0 ms    0 ms    10.131.0.200

Trace complete.

C:\>
```

Выводы по проделанной работе

Между сетью Университета г. Пиза и сетью «Донская» настроен GRE VPN-туннель. Для обмена маршрутной информацией использован OSPF.

Доступность узла сети Университета г. Пиза проверена с ноутбука администратора сети «Донская» с помощью команд ping и tracert.